Handling Imbalanced Data in Binary Classification: A Review and Comparative Experimental Study

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Abstract

A clear and well-documented LaTeX document is presented as an article formatted for publication by ACM in a conference proceedings or journal publication. Based on the "acmart" document class, this article presents and explains many of the common variations, as well as many of the formatting elements an author may use in the preparation of the documentation of their work.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

1 Introduction

1.1 Background and Problem Definition

In recent years, the rapid expansion of data availability across large-scale, complex, and networked systems has propelled machine learning to prominent roles in numerous application domains [1].

However, real-world data distributions are inherently complex and frequently exhibit a significant class imbalance. The class imbalance problem arises when the number of instances belonging to one class (the majority class) substantially outnumbers those of another (the minority class), resulting in a severely skewed training distribution [1, 2].

This phenomenon is pervasive in many high-stakes domains, including financial fraud detection, network intrusion detection, and the diagnosis of rare diseases such as diabetes and stroke, where the minority class—despite its low frequency of occurrence—typically constitutes the primary target of predictive interest [3, 4].

1.2 Challenges and the Pitfalls of Evaluation Metrics

Standard machine learning models are generally designed under the implicit assumption of an approximately balanced class distribution. Consequently, when applied to skewed data, these models exhibit a systematic bias toward the majority class, yielding poor predictive performance for minority instances—precisely those of greatest practical concern [1].

This issue is further compounded by the widespread use of conventional evaluation metrics. Overall accuracy, in particular, is highly misleading under class imbalance: a naive classifier that assigns all instances to the majority class can attain superficially high accuracy while entirely failing to identify any minority-class example [5].

To overcome this limitation, the research community has advocated the adoption of more discriminative metrics, including the F1-Score, the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) [6], and the Geometric Mean (G-Mean) [7], which collectively provide a more faithful assessment of model performance across both classes.

1.3 Taxonomy of Imbalanced Learning Techniques

To address the challenges posed by imbalanced datasets, a rich body of literature has proposed numerous techniques, which are broadly categorized into three main families: data-level methods, algorithmic-level methods, and hybrid strategies [1, 8].

Data-level approaches modify the training data distribution prior to model induction, either by oversampling the minority class—most notably through the Synthetic Minority Over-sampling Technique (SMOTE) [2] and its variants—or by undersampling the majority class through methods such as Random Under-Sampling (RUS).

Algorithmic-level approaches, by contrast, leave the data distribution unchanged and instead modify the learning procedure itself; representative techniques include cost-sensitive learning and ensemble-based methods such as weighted classifiers and boosting frameworks [9].

Finally, hybrid strategies combine the complementary strengths of both data resampling and algorithmic robustness, with methods such as SMOTEBoost and RUSBoost serving as prominent examples [9].

1.4 Research Gap and Objectives

Despite the substantial volume of existing work, many comparative studies evaluate imbalanced learning techniques across heterogeneous datasets that vary simultaneously in multiple dimensions—including class overlap, feature complexity, and sample size—making it difficult to isolate the independent effect of the Imbalance Ratio (IR) on classifier performance [1]. Confounding factors of this nature limit the interpretability of empirical findings and reduce their practical utility as guidelines for practitioners.

To address this gap, this study proposes a rigorous empirical framework grounded in controlled experimental design. Rather than relying solely on a diverse collection of independent datasets, we systematically vary the IR within a curated set of benchmark datasets spanning a wide range of imbalance severity—from mild (IR ≈ 2.4) to extreme (IR ≈ 41.4)—thereby enabling direct, controlled comparisons across escalating levels of class imbalance.

Within this framework, we conduct a systematic evaluation of representative methods from all three technique families—data-level, algorithmic-level, and hybrid—using three established base classifiers (SVM, Random Forest, and XGBoost) and a comprehensive set of evaluation metrics (F1-Score, AUC-ROC, and G-Mean). By analyzing the performance degradation trajectories of each strategy as class imbalance worsens, this study aims to provide both a nuanced empirical understanding of the strengths and limitations of existing approaches and actionable practical guidance for selecting appropriate imbalanced learning techniques in binary classification tasks.

2 Template Overview

As noted in the introduction, the "`acmart`" document class can be used to prepare many different kinds of documentation — a double-anonymous initial submission of a full-length technical paper, a two-page SIGGRAPH Emerging Technologies abstract, a "camera-ready" journal article, a SIGCHI Extended Abstract, and more — all by selecting the appropriate *template style* and *template parameters*.

This document will explain the major features of the document class. For further information, the *LaTeX User's Guide* is available from https://www.acm.org/publications/proceedings-template.

2.1 Template Styles

The primary parameter given to the "`acmart`" document class is the *template style* which corresponds to the kind of publication or SIG publishing the work. This parameter is enclosed in square brackets and is a part of the `documentclass` command:

```
\documentclass[STYLE]{acmart}
```

Journals use one of three template styles. All but three ACM journals use the `acmsmall` template style:

- `acmsmall`: The default journal template style.
- `acmlarge`: Used by JOCCH and TAP.
- `acmtog`: Used by TOG.

The majority of conference proceedings documentation will use the `acmconf` template style.

- `sigconf`: The default proceedings template style.
- `sigchi`: Used for SIGCHI conference articles.
- `sigplan`: Used for SIGPLAN conference articles.

2.2 Template Parameters

In addition to specifying the *template style* to be used in formatting your work, there are a number of *template parameters* which modify some part of the applied template style. A complete list of these parameters can be found in the *LaTeX User's Guide*.

Frequently-used parameters, or combinations of parameters, include:

- `anonymous,review`: Suitable for a "double-anonymous" conference submission. Anonymizes the work and includes line numbers. Use with the `\acmSubmissionID` command to print the submission's unique ID on each page of the work.
- `authorversion`: Produces a version of the work suitable for posting by the author.
- `screen`: Produces colored hyperlinks.

This document uses the following string as the first command in the source file:

```
\documentclass[sigconf]{acmart}
```

3 EXPERIMENTAL SETUP

3.1 Datasets

To rigorously investigate the effect of varying Imbalance Ratios (IR) on classifier performance, this study employs eight benchmark datasets sourced from the KEEL (Knowledge Extraction based on Evolutionary Learning) repository [?], a widely adopted resource for imbalanced classification research. The selected datasets span a broad spectrum of imbalance severity—from mild to extreme—and originate from diverse real-world domains, thereby enabling controlled cross-IR comparisons while maintaining ecological validity.

The Imbalance Ratio is defined formally as:

$$\mathrm{IR} = \frac{|S_{\mathrm{maj}}|}{|S_{\mathrm{min}}|}, \tag{1}$$

where $|S_{\mathrm{maj}}|$ and $|S_{\mathrm{min}}|$ denote the cardinalities of the majority and minority classes, respectively. Following the severity categorization proposed in the imbalanced learning literature [?], the datasets are stratified into four levels: *mild* ($\mathrm{IR} < 3$), *moderate* ($3 \leq \mathrm{IR} < 15$), *severe* ($15 \leq \mathrm{IR} < 30$), and *extreme* ($\mathrm{IR} \geq 30$).

All experiments employ **stratified 5-fold cross-validation** to ensure that the class distribution is preserved in every fold. The complete characteristics of the benchmark suite are summarised in Table 1.

The inclusion of four Yeast variants (Yeast-1, -3, -4, and -6) sharing an identical feature space and sample size ($n = 1{,}484$) is a deliberate methodological choice: by holding feature complexity and data size constant while varying only the IR, these four datasets allow the direct isolation of class imbalance severity as the sole experimental variable, effectively controlling for confounds that commonly hinder cross-dataset comparisons [?].

3.2 Evaluation Metrics

A central challenge in evaluating classifiers on imbalanced datasets is metric selection. As demonstrated by He and Garcia [?] and mathematically formalised by Luque et al. [?], overall *Accuracy* is a fundamentally flawed metric under class imbalance: a degenerate classifier that assigns every instance to the majority class can attain superficially high accuracy while entirely failing to detect any minority-class example. Accordingly, this study excludes Accuracy as a primary performance measure and instead adopts a suite of imbalance-aware metrics grounded in the binary confusion matrix, whose elements are defined as follows: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), where the *positive* class denotes the minority class throughout.

Precision and Recall. Precision measures the proportion of predicted positive instances that are truly positive; Recall (Sensitivity) measures the proportion of actual positive instances that are correctly identified. Both metrics are defined over the minority class and are expressed as [?]:

$$\mathrm{Precision} = \frac{TP}{TP + FP}, \tag{2}$$

Table 1: Summary of Benchmark Datasets.

Dataset	Instances	Features	Minority	Majority	Imbalance Ratio	Severity
Yeast-1	1,484	8	429	1,055	2.46	Mild
Yeast-3	1,484	8	163	1,321	8.10	Moderate
Yeast-4	1,484	8	51	1,433	28.10	Severe
Yeast-6	1,484	8	35	1,449	41.40	Extreme

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{3}$$

F1-Score. The F1-Score is the harmonic mean of Precision and Recall, providing a single scalar that penalises severe imbalances between the two [?]:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\,TP}{2\,TP + FP + FN}. \tag{4}$$

Because the harmonic mean is strictly lower than the arithmetic mean whenever Precision and Recall diverge, F_1 constitutes a conservative and informative composite measure for imbalanced classification tasks.

AUC-ROC.. The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is a threshold-independent metric that quantifies a classifier's ability to discriminate between classes across all possible decision thresholds [?]. Let the False Positive Rate (FPR) and True Positive Rate (TPR) be defined as:

$$\text{FPR} = \frac{FP}{FP + TN}, \qquad \text{TPR} = \frac{TP}{TP + FN}. \tag{5}$$

The AUC-ROC is then computed as:

$$\text{AUC}_{\text{ROC}} = \int_0^1 \text{TPR}\big(\text{FPR}^{-1}(x)\big)\, dx. \tag{6}$$

A value of 0.5 corresponds to random classification, while 1.0 denotes a perfect classifier. As noted by Saito and Rehmsmeier [?], however, ROC-based metrics can present an overly optimistic picture under extreme class imbalance, because the large pool of true negatives in the majority class suppresses the FPR, thereby inflating the apparent curve.

G-Mean. The Geometric Mean (G-Mean), originally proposed by Kubat and Matwin [?] for imbalanced learning, is the square root of the product of Sensitivity (Recall) and Specificity. It provides a balanced assessment of performance across both classes simultaneously: poor performance on *either* class leads to a sharp decline in the overall score. Specificity and G-Mean are defined as:

$$\text{Specificity} = \frac{TN}{TN + FP}, \tag{7}$$

$$\text{G-Mean} = \sqrt{\text{Recall} \times \text{Specificity}} = \sqrt{\frac{TP}{TP + FN} \cdot \frac{TN}{TN + FP}}. \tag{8}$$

Metric Selection Rationale. Although this study reports all metrics described above for completeness, the **primary evaluation metrics are F1-Score, AUC-ROC, and G-Mean**. These three are selected because they collectively capture complementary dimensions of classifier performance under imbalance: the F1-Score reflects the precision–recall trade-off for the minority class; the AUC-ROC captures threshold-independent ranking ability across all decision boundaries; and the G-Mean ensures that majority-class performance is not sacrificed for minority-class gains. Metrics including overall Accuracy and Balanced Accuracy (BAcc) are reported as supplementary references but are not used as the basis for primary comparisons, in accordance with the limitations identified in prior literature [? ? ?].

4 Typefaces

The "`acmart`" document class requires the use of the "Libertine" typeface family. Your TEX installation should include this set of packages. Please do not substitute other typefaces. The "`lmodern`" and "`ltimes`" packages should not be used, as they will override the built-in typeface families.

5 Title Information

The title of your work should use capital letters appropriately - https://capitalizemytitle.com/ has useful rules for capitalization. Use the `title` command to define the title of your work. If your work has a subtitle, define it with the `subtitle` command. Do not insert line breaks in your title.

If your title is lengthy, you must define a short version to be used in the page headers, to prevent overlapping text. The `title` command has a "short title" parameter:

```
\title[short title]{full title}
```

6 Authors and Affiliations

Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors' names should not be abbreviated; use full first names wherever possible. Include authors' e-mail addresses whenever possible.

Grouping authors' names or e-mail addresses, or providing an "e-mail alias," as shown below, is not acceptable:

```
\author{Brooke Aster, David Mehldau}
\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

The `authornote` and `authornotemark` commands allow a note to apply to multiple authors — for example, if the first two authors of an article contributed equally to the work.

If your author list is lengthy, you must define a shortened version of the list of authors to be used in the page headers, to prevent overlapping text. The following command should be placed just after the last `\author{}` definition:

```
\renewcommand{\shortauthors}{McCartney, et al.}
```

Omitting this command will force the use of a concatenated list of all of the authors' names, which may result in overlapping text in the page headers.

The article template's documentation, available at https://www.acm.org/publications/proceedings-template, has a complete explanation of these commands and tips for their effective use.

Note that authors' addresses are mandatory for journal articles.

7 Rights Information

Authors of any work published by ACM will need to complete a rights form. Depending on the kind of work, and the rights management choice made by the author, this may be copyright transfer, permission, license, or an OA (open access) agreement.

Regardless of the rights management choice, the author will receive a copy of the completed rights form once it has been submitted. This form contains LaTeX commands that must be copied into the source document. When the document source is compiled, these commands and their parameters add formatted text to several areas of the final document:

- the "ACM Reference Format" text on the first page.
- the "rights management" text on the first page.
- the conference information in the page header(s).

Rights information is unique to the work; if you are preparing several works for an event, make sure to use the correct set of commands with each of the works.

The ACM Reference Format text is required for all articles over one page in length, and is optional for one-page articles (abstracts).

8 CCS Concepts and User-Defined Keywords

Two elements of the "acmart" document class provide powerful taxonomic tools for you to help readers find your work in an online search.

The ACM Computing Classification System — https://www.acm.org/publications/class-2012 — is a set of classifiers and concepts that describe the computing discipline. Authors can select entries from this classification system, via https://dl.acm.org/ccs/ccs.cfm, and generate the commands to be included in the LaTeX source.

User-defined keywords are a comma-separated list of words and phrases of the authors' choosing, providing a more flexible way of describing the research being presented.

CCS concepts and user-defined keywords are required for for all articles over two pages in length, and are optional for one- and two-page articles (or abstracts).

9 Sectioning Commands

Your work should use standard LaTeX sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsubsection` should be numbered; do not remove the numbering from the commands.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed.**

Below are examples of sectioning commands.

9.1 Subsection

This is a subsection.

Table 2: Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

9.1.1 *Subsubsection.* This is a subsubsection.

Paragraph. This is a paragraph.
Subparagraph This is a subparagraph.

10 Tables

The "`acmart`" document class includes the "`booktabs`" package — https://ctan.org/pkg/booktabs — for preparing high-quality tables.

Table captions are placed *above* the table.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper "floating" placement of tables, use the environment **table** to enclose the table's contents and the table caption. The contents of the table itself must go in the **tabular** environment, to be aligned properly in rows and columns, with the desired horizontal and vertical rules. Again, detailed instructions on **tabular** material are found in the *LaTeX User's Guide*.

Immediately following this sentence is the point at which Table 2 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page's live area, use the environment **table*** to enclose the table's contents and the table caption. As with a single-column table, this wide table will "float" to a location deemed more desirable. Immediately following this sentence is the point at which Table 3 is included in the input file; again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

Always use midrule to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily.

11 Math Equations

You may want to display math equations in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections.

11.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the **math** environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. You can use any of the symbols and structures, from α to ω, available in LaTeX [25]; this section will simply show a few examples of in-text equations in context. Notice how this equation: $\lim_{n\rightarrow \infty}x=0$, set here in in-line math style, looks slightly different when set in display style. (See next section).

Table 3: Some Typical Commands

Command	A Number	Comments
`\author`	100	Author
`\table`	300	For tables
`\table*`	400	For wider tables

11.2 Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the **equation** environment. An unnumbered display equation is produced by the **displaymath** environment.

Again, in either environment, you can use any of the symbols and structures available in LaTeX; this section will just give a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n\rightarrow \infty}x=0 \tag{9}$$

Notice how it is formatted somewhat differently in the **displaymath** environment. Now, we'll enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

and follow it with another numbered equation:

$$\sum_{i=0}^{\infty}x_i=\int_{0}^{\pi+2} f \tag{10}$$

just to demonstrate LaTeX's able handling of numbering.

12 Figures

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Your figures should contain a caption which describes the figure to the reader.

Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption - their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see https://www.acm.org/publications/taps/describing-figures/.

12.1 The "Teaser Figure"

A "teaser figure" is an image, or set of images in one figure, that are placed after all author and affiliation information, and before the body of the article, spanning the page. If you wish to have such a figure in your article, place the command immediately before the `\maketitle` command:

```
\begin{teaserfigure}
  \includegraphics[width=\textwidth]{sampleteaser}
  \caption{figure caption}
  \Description{figure description}
\end{teaserfigure}
```

13 Citations and Bibliographies

The use of BibTeX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete — use full first names ("Donald E. Knuth") not initials ("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

where "`bibfile`" is the name, without the "`.bib`" suffix, of the BibTeX file.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the "author year" style; for these exceptions, please include this command in the **preamble** (before the command "`\begin{document}`") of your LaTeX source:

```
\citestyle{acmauthoryear}
```

Some examples. A paginated journal article [2], an enumerated journal article [11], a reference to an entire issue [10], a monograph (whole book) [24], a monograph/whole book in a series (see 2a in spec. document) [18], a divisible-book such as an anthology or compilation [13] followed by the same example, however we only output the series if the volume number is given [14] (so Editor00a's series should NOT be present since it has no vol. no.), a chapter in a divisible book [37], a chapter in a divisible book in a series [12], a multi-volume work as book [23], a couple of articles in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [3, 16], a proceedings article with all possible elements [36], an example of an enumerated proceedings article [15], an informally published work [17], a couple of preprints [6, 8], a doctoral dissertation [9], a master's thesis: [4], an online document / world wide web resource [1, 29, 38], a video game (Case 1) [28] and (Case 2) [27] and [26] and (Case 3) a patent [35], work

accepted for publication [32], 'YYYYb'-test for prolific author [33] and [34]. Other cites might contain 'duplicate' DOI and URLs (some SIAM articles) [22]. Boris / Barbara Beeton: multi-volume works as books [20] and [19]. A presentation [31]. An article under review [7]. A couple of citations with DOIs: [21, 22]. Online citations: [38–40]. Artifacts: [30] and [5].

14 Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document.

This section has a special environment:

```
\begin{acks}
...
\end{acks}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading.

Authors should not prepare this section as a numbered or unnumbered `\section`; please use the "`acks`" environment.

15 Appendices

If your work needs an appendix, add it before the "`\end{document}`" command at the conclusion of your source document.

Start the appendix with the "`appendix`" command:

```
\appendix
```

and note that in the appendix, sections are lettered, not numbered. This document has two appendices, demonstrating the section and subsection identification method.

16 Multi-language papers

Papers may be written in languages other than English or include titles, subtitles, keywords and abstracts in different languages (as a rule, a paper in a language other than English should include an English title and an English abstract). Use `language=...` for every language used in the paper. The last language indicated is the main language of the paper. For example, a French paper with additional titles and abstracts in English and German may start with the following command

```
\documentclass[sigconf, language=english, language=german,
                language=french]{acmart}
```

The title, subtitle, keywords and abstract will be typeset in the main language of the paper. The commands `\translatedXXX`, `XXX` begin title, subtitle and keywords, can be used to set these elements in the other languages. The environment `translatedabstract` is used to set the translation of the abstract. These commands and environment have a mandatory first argument: the language of the second argument. See `sample-sigconf-i13n.tex` file for examples of their usage.

17 SIGCHI Extended Abstracts

The "`sigchi-a`" template style (available only in LaTeX and not in Word) produces a landscape-orientation formatted article, with a wide left margin. Three environments are available for use with the "`sigchi-a`" template style, and produce formatted output in the margin:

`sidebar:` Place formatted text in the margin.
`marginfigure:` Place a figure in the margin.
`margintable:` Place a table in the margin.

Acknowledgments

To Robert, for the bagels and explaining CMYK and color spaces.

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A Research Methods

A.1 Part One

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C Experimental Results Summary

This section provides a comprehensive summary of the experimental results across all evaluated methods and datasets.

D Detailed Experimental Analysis

In this section, we provide a detailed analysis of the experimental results, categorized by the level of intervention: data-level, algorithm-level, and hybrid strategies.

D.1 Data Level Methods

Data-level methods focus on reslicing the training set to achieve a more balanced distribution. This section presenting the G-Mean performance for individual classifiers and visual analysis of the results.

D.1.1 Experimental Tables: Data-Level Results.

D.1.2 Visual Analysis of Data-Level Methods. This subcategory compares the G-mean performance across multiple classifiers (KNN, Random Forest, SVM) using different data-resampling techniques. Figures 1 through 4 illustrate the impact of oversampling, SMOTE, and SMOTE+ENN.

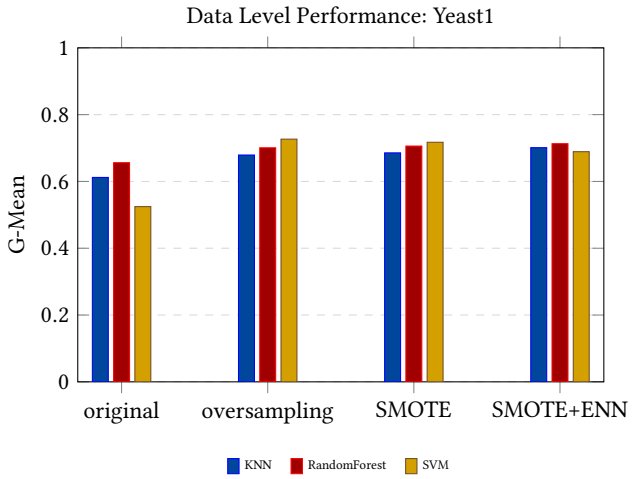

Figure 1: [Data Level] G-Mean performance comparison on Yeast1 dataset.

D.2 Algorithm Level Methods

Algorithm-level methods, such as Cost-Sensitive learning combined with ensemble classifiers (Random Forest, XGBoost, AdaBoost),

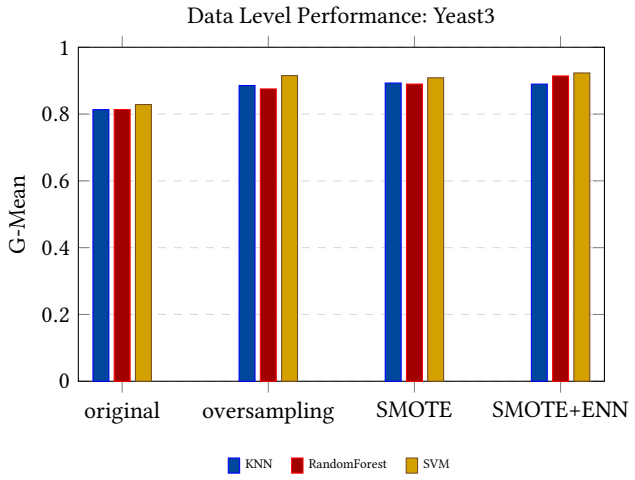

Figure 2: [Data Level] G-Mean performance comparison on Yeast3 dataset.

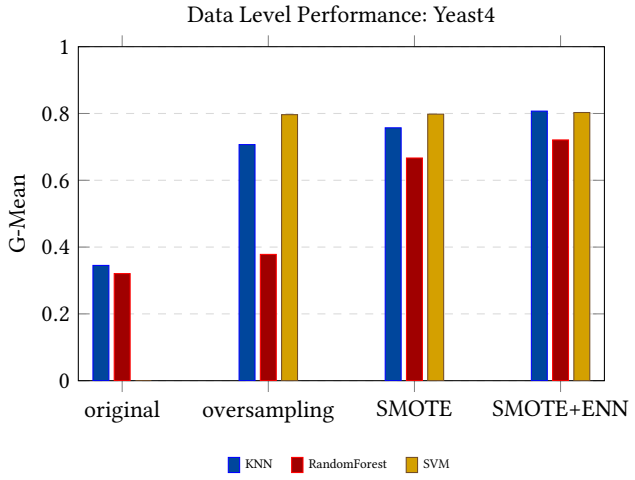

Figure 3: [Data Level] G-Mean performance comparison on Yeast4 dataset.

show higher robustness across varying imbalance ratios by directly internalizing the cost of misclassifying minority instances.

D.2.1 Visual Analysis of Algorithm-Level Methods. In this subcategory, we examine the performance shifts in ensemble methods when cost-sensitive mechanisms are applied.

Figure 5 presents a comparison of Recall scores between standard Random Forest and Weighted (Cost-Sensitive) Random Forest across the Yeast datasets.

Furthermore, Table 4 and Figure 6 highlight the "death cross" effect in AdaBoost. After applying cost-sensitive weights, Recall improves significantly, but at a severe cost to Precision.

Finally, the radar chart in Figure 7 illustrates the balanced improvement across multiple metrics (Recall, Precision, F1-Score, and G-Mean) for XGBoost when cost-sensitive learning is integrated.

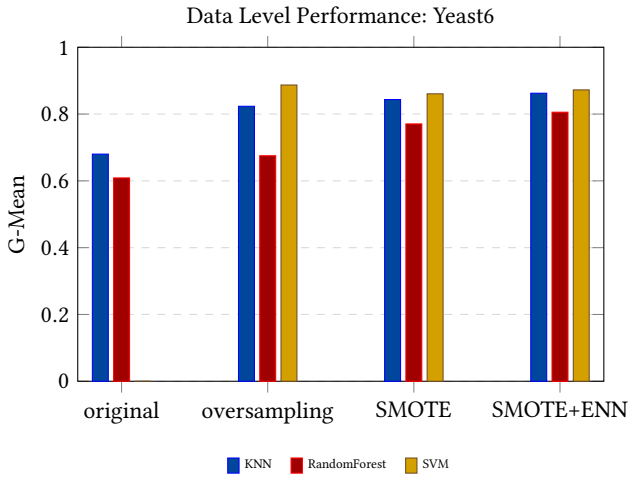

Figure 4: [Data Level] G-Mean performance comparison on Yeast6 dataset.

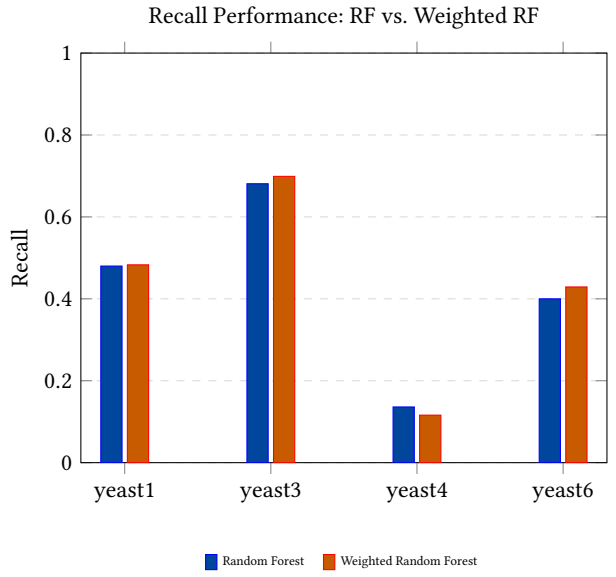

Figure 5: [Algorithm Level] Recall comparison of standard and weighted Random Forest.

D.3 Hybrid Methods

Hybrid methods integrate data resampling with algorithmic enhancements. This subsection evaluates the efficacy of boosting-based hybrid approaches.

D.3.1 Visual Analysis of Hybrid Methods. The hybrid framework combines data-level resampling with algorithm-level boosting. Figure 8 and Figure 9 provide insight into these combined effects.

Figure 8 illustrates the F1-score performance of various hybrid boosting methods across four Yeast datasets.

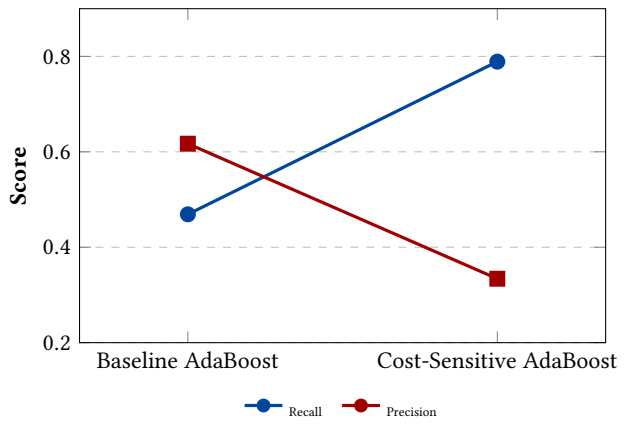

Figure 6: [Algorithm Level] Dual-metric line chart illustrating the trade-off in AdaBoost.

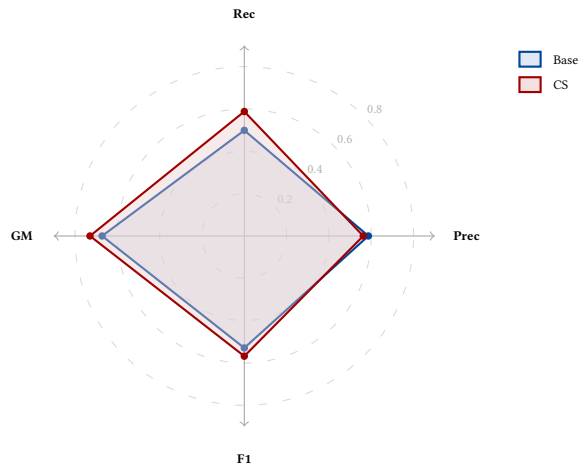

Figure 7: [Algorithm Level] Radar Chart comparing Average XGBoost vs. CS-XGBoost.

D.3.2 Impact of SMOTE Variants in Hybrid Frameworks. Figure 9 shows the comparative performance of SMOTE-based hybrid strategies. SMOTEBoost demonstrates superior F1-scores compared to the pure data-level SMOTE and the cost-sensitive variant SMOTECSL, particularly as the imbalance ratio increases in Yeast-4 and Yeast-6.

D.4 Summary Statement

The experimental results demonstrate that while data-level methods provide a foundational improvement, algorithm-level and hybrid approaches offer superior stability and performance across extreme imbalance ratios.

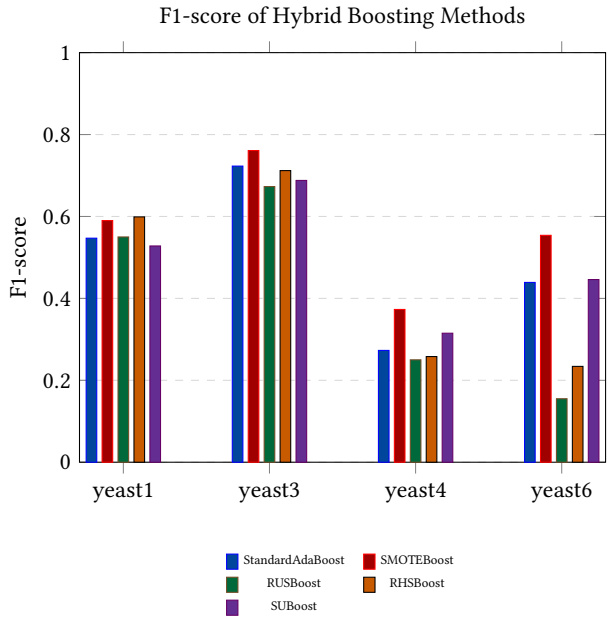

Figure 8: [Hybrid Method] Performance comparison of various hybrid boosting methods (based on exp_1.csv).

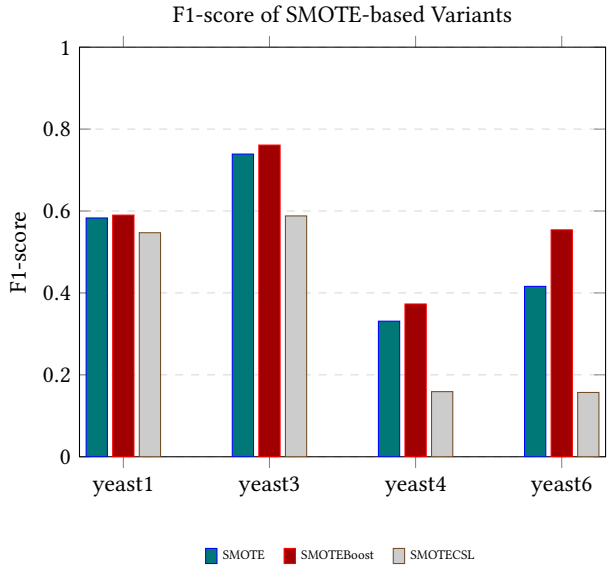

Figure 9: [Hybrid Method] Impact of SMOTE variants on model performance (based on exp_2.csv).

Table 4: Comprehensive Experimental Results Summary across all methodology levels.

Type	Method	Dataset	Acc.	Prec.	Rec.	F1	AUC	PR-AUC	G-Mean	BAcc.
Data Level	Baseline	yeast1	0.7581	0.6410	0.4025	0.4850	0.7881	0.5887	0.5976	0.6526
		yeast3	0.9477	0.8106	0.6848	0.7412	0.9570	0.7924	0.8184	0.8325
		yeast4	0.9661	0.3556	0.0970	0.1403	0.8561	0.3739	0.2219	0.5470
		yeast6	0.9798	0.4556	0.2952	0.3445	0.8999	0.5656	0.4296	0.6458
	oversampling	yeast1	0.7280	0.5356	0.6596	0.5841	0.7835	0.5737	0.7021	0.7077
		yeast3	0.9292	0.6479	0.8488	0.7290	0.9509	0.7530	0.8920	0.8940
		yeast4	0.9261	0.3308	0.4897	0.3022	0.8732	0.3626	0.6272	0.7157
		yeast6	0.9537	0.3796	0.6857	0.4355	0.9080	0.4895	0.7952	0.8230
	SMOTE	yeast1	0.7201	0.5209	0.6760	0.5831	0.7834	0.5747	0.7029	0.7070
		yeast3	0.9328	0.6561	0.8551	0.7391	0.9571	0.7664	0.8971	0.8988
		yeast4	0.9115	0.2465	0.6152	0.3315	0.8801	0.3527	0.7406	0.7686
		yeast6	0.9452	0.3202	0.7333	0.4155	0.9240	0.4788	0.8248	0.8418
	SMOTE-ENN	yeast1	0.6734	0.4657	0.7948	0.5846	0.7803	0.5666	0.7011	0.7094
		yeast3	0.9216	0.6028	0.8938	0.7177	0.9533	0.7438	0.9089	0.9094
		yeast4	0.8821	0.1900	0.6921	0.2928	0.8881	0.3371	0.7769	0.7905
		yeast6	0.9306	0.2587	0.7810	0.3724	0.9324	0.4875	0.8468	0.8576
Algorithm Level	RandomForest	yeast1	0.7783	0.6650	0.4803	0.5553	0.8133	0.6435	0.6561	0.6899
		yeast3	0.9447	0.7870	0.6805	0.7275	0.9706	0.8316	0.8134	0.8289
		yeast4	0.9677	0.6000	0.1364	0.2062	0.9206	0.4073	0.3206	0.5668
		yeast6	0.9831	0.7667	0.4000	0.5054	0.8997	0.5629	0.6088	0.6986
	CS-RandomForest	yeast1	0.7891	0.6979	0.4825	0.5688	0.8114	0.6420	0.6632	0.6981
		yeast3	0.9461	0.7896	0.6992	0.7396	0.9726	0.8479	0.8254	0.8379
		yeast4	0.9650	0.3333	0.1164	0.1655	0.9269	0.4298	0.2572	0.5557
		yeast6	0.9838	0.8667	0.4286	0.5305	0.8965	0.5907	0.6350	0.7129
	XGBoost	yeast1	0.7648	0.6122	0.5151	0.5569	0.7847	0.6034	0.6662	0.6907
		yeast3	0.9434	0.7567	0.7176	0.7354	0.9612	0.7726	0.8338	0.8444
		yeast4	0.9629	0.4503	0.2764	0.3241	0.9165	0.3767	0.5024	0.6319
		yeast6	0.9771	0.5267	0.4857	0.4992	0.9246	0.5340	0.6875	0.7373
	CS-XGBoost	yeast1	0.7561	0.5765	0.5989	0.5861	0.7859	0.6012	0.6998	0.7094
		yeast3	0.9468	0.7482	0.7852	0.7649	0.9615	0.7895	0.8709	0.8760
		yeast4	0.9582	0.4230	0.3945	0.3932	0.9107	0.4053	0.6066	0.6865
		yeast6	0.9764	0.4984	0.5714	0.5294	0.9242	0.5515	0.7399	0.7788
	AdaBoost (d=1)	yeast1	0.7601	0.6293	0.4219	0.5045	0.7878	0.5760	0.6152	0.6598
		yeast3	0.9508	0.7891	0.7725	0.7770	0.9721	0.7808	0.8664	0.8726
		yeast4	0.9623	0.3500	0.1945	0.2439	0.9217	0.3758	0.3814	0.5920
		yeast6	0.9831	0.7000	0.4857	0.5655	0.9380	0.5549	0.6836	0.7404
	CS-AdaBoost	yeast1	0.6658	0.4496	0.6922	0.5449	0.6736	0.4009	0.6729	0.6736
		yeast3	0.8956	0.5169	0.9203	0.6610	0.9470	0.7058	0.9061	0.9064
		yeast4	0.8922	0.2093	0.7436	0.3251	0.9190	0.2352	0.8154	0.8205
		yeast6	0.8956	0.1608	0.8000	0.2655	0.8972	0.2441	0.8374	0.8489
Hybrid Method	AdaBoost (d=5)	yeast1	0.7540	0.5853	0.5196	0.5470	0.7821	0.5880	0.6612	0.6845
		yeast3	0.9421	0.7577	0.6928	0.7231	0.9591	0.7695	0.8199	0.8328
		yeast4	0.9650	0.5500	0.1964	0.2727	0.9312	0.4115	0.3923	0.5943
		yeast6	0.9778	0.5500	0.3714	0.4392	0.8979	0.4859	0.6043	0.6819
	SMOTEBoost	yeast1	0.7473	0.5560	0.6294	0.5903	0.7718	0.5543	0.7074	0.7123
		yeast3	0.9468	0.7530	0.7725	0.7611	0.9596	0.7989	0.8641	0.8704
		yeast4	0.9535	0.3799	0.3946	0.3730	0.9147	0.4122	0.6148	0.6840
		yeast6	0.9778	0.5101	0.6286	0.5539	0.9147	0.5500	0.7704	0.8074
	RUSBoost	yeast1	0.7156	0.5171	0.6085	0.5498	0.7558	0.5147	0.6728	0.6839
		yeast3	0.9003	0.5325	0.9203	0.6731	0.9348	0.5969	0.9090	0.9090
		yeast4	0.8255	0.1469	0.8418	0.2499	0.8847	0.1846	0.8321	0.8333
		yeast6	0.7864	0.0856	0.8286	0.1548	0.8265	0.0983	0.8009	0.8070
	RHSBoost	yeast1	0.7507	0.5613	0.6456	0.5993	0.7910	0.5896	0.7148	0.7195
		yeast3	0.9191	0.5901	0.9015	0.7115	0.9546	0.6852	0.9111	0.9114
		yeast4	0.8396	0.1545	0.8036	0.2584	0.8824	0.1807	0.8210	0.8223
		yeast6	0.8679	0.1359	0.8571	0.2340	0.8675	0.1328	0.8580	0.8627
	SUBoost	yeast1	0.7224	0.5297	0.5408	0.5277	0.7571	0.5586	0.6512	0.6685
		yeast3	0.9394	0.7743	0.6267	0.6877	0.9548	0.7286	0.7778	0.8024
		yeast4	0.9589	0.4417	0.2746	0.3146	0.8879	0.3961	0.4581	0.6289
		yeast6	0.9771	0.5343	0.4286	0.4462	0.8706	0.4566	0.6227	0.7095
	SMOTECSL	yeast1	0.5492	0.3855	0.9416	0.5470	0.7931	0.5956	0.6053	0.6656
		yeast3	0.8538	0.4260	0.9510	0.5883	0.9688	0.8093	0.8947	0.8964
		yeast4	0.6874	0.0877	0.8618	0.1591	0.8535	0.1488	0.7648	0.7715
		yeast6	0.7540	0.0865	0.9429	0.1573	0.9100	0.3574	0.8363	0.8462

Table 5: [Data Level] Summary of Experimental Results (Data-Level Methods)

Method	Dataset	Model	Acc.	Prec.	Rec.	F1	AUC	PR-AUC	G-Mean	BAcc
Original	yeast1	KNN	0.7392	0.5623	0.4360	0.4904	0.7550	0.5083	0.6120	0.6492
		RF	0.7783	0.6650	0.4800	0.5553	0.8133	0.6435	0.6561	0.6899
		SVM	0.7568	0.6956	0.2910	0.4092	0.7961	0.6142	0.5246	0.6187
	yeast3	KNN	0.9468	0.8119	0.6750	0.7362	0.9249	0.7188	0.8133	0.8277
		RF	0.9447	0.7870	0.6800	0.7275	0.9706	0.8316	0.8134	0.8289
		SVM	0.9515	0.8330	0.6990	0.7598	0.9755	0.8268	0.8285	0.8408
	yeast4	KNN	0.9650	0.4667	0.1550	0.2147	0.7944	0.2928	0.3451	0.5741
		RF	0.9677	0.6000	0.1360	0.2062	0.9206	0.4073	0.3206	0.5668
		SVM	0.9656	0.0000	0.0000	0.0000	0.8532	0.4215	0.0000	0.5000
	yeast6	KNN	0.9798	0.6000	0.4860	0.5281	0.8997	0.5167	0.6801	0.7387
		RF	0.9831	0.7667	0.4000	0.5054	0.8997	0.5629	0.6088	0.6986
		SVM	0.9764	0.0000	0.0000	0.0000	0.9003	0.6173	0.0000	0.5000
Oversampling	yeast1	KNN	0.6894	0.4727	0.6573	0.5498	0.7382	0.4774	0.6791	0.6799
		RF	0.7763	0.6244	0.5734	0.5964	0.8120	0.6383	0.7007	0.7161
		SVM	0.7183	0.5098	0.7481	0.6061	0.8003	0.6054	0.7266	0.7271
	yeast3	KNN	0.9131	0.5730	0.8525	0.6838	0.9151	0.6105	0.8855	0.8865
		RF	0.9488	0.7556	0.7917	0.7729	0.9666	0.8332	0.8752	0.8799
		SVM	0.9259	0.6151	0.9023	0.7303	0.9709	0.8153	0.9153	0.9156
	yeast4	KNN	0.9360	0.2700	0.5491	0.3585	0.7934	0.2407	0.7068	0.7494
		RF	0.9663	0.5444	0.1964	0.2624	0.9276	0.4124	0.3781	0.5950
		SVM	0.8760	0.1779	0.7236	0.2856	0.8987	0.4346	0.7965	0.8025
	yeast6	KNN	0.9589	0.3262	0.7143	0.4477	0.8954	0.3599	0.8233	0.8395
		RF	0.9805	0.5993	0.4857	0.5177	0.8890	0.5812	0.6752	0.7391
		SVM	0.9218	0.2134	0.8571	0.3411	0.9397	0.5274	0.8870	0.8903
SMOTE	yeast1	KNN	0.6934	0.4783	0.6691	0.5575	0.7447	0.4872	0.6855	0.6862
		RF	0.7642	0.5911	0.6013	0.5952	0.8043	0.6297	0.7059	0.7158
		SVM	0.7028	0.4932	0.7575	0.5965	0.8011	0.6070	0.7173	0.7190
	yeast3	KNN	0.9164	0.5817	0.8648	0.6944	0.9330	0.6695	0.8929	0.8938
		RF	0.9481	0.7363	0.8223	0.7766	0.9666	0.8186	0.8898	0.8930
		SVM	0.9340	0.6502	0.8780	0.7463	0.9715	0.8111	0.9086	0.9095
	yeast4	KNN	0.9036	0.2039	0.6473	0.3077	0.8303	0.2104	0.7571	0.7800
		RF	0.9522	0.3516	0.4745	0.3938	0.9107	0.3857	0.6667	0.7219
		SVM	0.8787	0.1839	0.7236	0.2929	0.8993	0.4619	0.7980	0.8039
	yeast6	KNN	0.9265	0.2144	0.7714	0.3344	0.9093	0.3032	0.8434	0.8509
		RF	0.9778	0.5167	0.6286	0.5573	0.9183	0.5391	0.7705	0.8074
		SVM	0.9313	0.2293	0.8000	0.3549	0.9444	0.5942	0.8606	0.8672
SMOTE+ENN	yeast1	KNN	0.6671	0.4570	0.8042	0.5828	0.7481	0.4748	0.7011	0.7078
		RF	0.7109	0.5007	0.7224	0.5903	0.7973	0.6146	0.7130	0.7143
		SVM	0.6422	0.4392	0.8577	0.5807	0.7954	0.6105	0.6891	0.7061
	yeast3	KNN	0.9003	0.5288	0.8771	0.6590	0.9222	0.6136	0.8897	0.8901
		RF	0.9394	0.6726	0.8837	0.7628	0.9666	0.8096	0.9141	0.9150
		SVM	0.9252	0.6070	0.9206	0.7312	0.9712	0.8083	0.9231	0.9232
	yeast4	KNN	0.8760	0.1826	0.7455	0.2929	0.8424	0.1908	0.8071	0.8131
		RF	0.9205	0.2305	0.5691	0.3253	0.9268	0.3933	0.7208	0.7511
		SVM	0.8497	0.1570	0.7618	0.2600	0.8951	0.4273	0.8029	0.8073
	yeast6	KNN	0.9036	0.1788	0.8286	0.2925	0.9077	0.2530	0.8624	0.8670
		RF	0.9676	0.3899	0.6857	0.4941	0.9457	0.5976	0.8056	0.8301
		SVM	0.9205	0.2073	0.8286	0.3305	0.9439	0.6119	0.8724	0.8756

Table 6: [Data Level] G-Mean Performance Comparison for KNN Classifier

Method	Yeast1	Yeast3	Yeast4	Yeast6
Original	0.6120	0.8133	0.3451	0.6801
Oversampling	0.6791	0.8855	0.7068	0.8233
SMOTE	0.6855	0.8929	0.7571	0.8434
SMOTE+ENN	0.7011	0.8897	0.8071	0.8624

Table 7: [Data Level] G-Mean Performance Comparison for Random Forest Classifier

Method	Yeast1	Yeast3	Yeast4	Yeast6
Original	0.6561	0.8134	0.3206	0.6088
Oversampling	0.7007	0.8752	0.3781	0.6752
SMOTE	0.7059	0.8898	0.6667	0.7705
SMOTE+ENN	0.7130	0.9141	0.7208	0.8056

Table 8: [Data Level] G-Mean Performance Comparison for SVM Classifier

Method	Yeast1	Yeast3	Yeast4	Yeast6
Original	0.5246	0.8285	0.0000	0.0000
Oversampling	0.7266	0.9153	0.7965	0.8870
SMOTE	0.7173	0.9086	0.7980	0.8606
SMOTE+ENN	0.6891	0.9231	0.8029	0.8724

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